10–12 Minute Presentation Slide Deck: AI-Augmented Security Analysis & Portfolio Construction

Project: Narrative vs. Numbers: An AI-Augmented Security Analysis of Trent, Asian Paints, and TCS
Course: Security Analysis & Portfolio Management | **Faculty:** Prof. Jagriti
Target Duration: 10–12 Minutes (45–50 seconds per slide)

Slide 1: Title & Group Details (0:00 – 0:45)

- **Title:** Narrative vs. Numbers: An AI-Augmented Security Analysis of Trent Ltd., Asian Paints Ltd., and Tata Consultancy Services (TCS), with a Lifecycle Portfolio Built on Utility-Based Allocation
- **Subtitle:** Where Market Sentiment Diverges from Corporate Fundamentals in Listed Indian Equities
- **Group Details:** [Student Name / Roll Number] | SAPM Course | Date: November 2026
- **Live Artifacts:** 16-Sheet Audited Master Excel Workbook (`SA_final_workbook.xlsx`) & **Folio (`F.`)** Dynamic TradingView + Ollama AI Terminal (`http://localhost:8502`)

Slide 2: The Core Question & Central Thesis (0:45 – 1:30)

- **Central Question:** Where does public market sentiment diverge from intrinsic valuation, and can that divergence be quantitatively measured, validated with NLP models, and exploited in multi-asset portfolio construction?
- **The Three Archetypes:**
    1. **Trent Ltd. (Hype / Momentum):** P/E multiple (94.1x) running far ahead of fundamental DCF cash flows based on store expansion enthusiasm.
    2. **Asian Paints Ltd. (De-Rating):** Multiples contracted sharply from 5Y median (61.2x to 54.1x) post Grasim Birla Opus entry as sentiment extrapolated temporary margin wars into permanent moat impairment.
    3. **TCS Ltd. (Fear vs. Fundamentals):** P/E compressed to 18.4x by Generative AI disruption fears, directly conflicting with 1.01x 5Y CFO/PAT conversion, Rs 72,398 Cr EBITDA, and >45% ROCE.

- **Core Takeaway:** AI is not an add-on; it directly tests and contextualizes the traditional valuation gaps.

Slide 3: Company Selection & Sector Diversification Proof (1:30 – 2:15)

- **Sector Diversification:** Retail (Trent), Decoratives/Chemicals (Asian Paints), Enterprise IT (TCS) — distinct economic demand drivers.
- **Empirical 5-Year Monthly Correlation Proof (Oct 2021 – Oct 2026):**
  - Asian Paints vs. TCS correlation: **0.185** (Near zero cross-correlation)
  - Trent vs. TCS correlation: **0.350 |** Trent vs. Asian Paints: **0.403**
  - Proves statistically that blending these 3 assets significantly lowers aggregate portfolio variance.
- **Explicit Alternatives Rejected:**
  - *ITC:* Conglomerate tax distortion (cigarette excise) and hotel demerger complexity.
  - *HUL / Nestle:* Redundant urban consumer FMCG demand overlapping with Asian Paints.
  - *Coal India:* Sovereign ownership and regulated pricing suppress market sentiment signals.

Slide 4: Traditional Security Analysis Summary (2:15 – 3:00)

- **Five-Year Operational Review (FY22 – FY26):**
  - *Trent:* Revenue grew from Rs 4,498 Cr to Rs 20,074 Cr (45.3% CAGR); EBITDA margin expanded from 12.8% to 18.7%; 5Y CFO/PAT = 1.19x.
  - *Asian Paints:* Steady 5.2% CAGR (Rs 29,101 Cr to Rs 35,584 Cr); margin pressure under Birla Opus competition; 5Y CFO/PAT = 1.11x.
  - *TCS:* Revenue grew from Rs 1,91,754 Cr to Rs 2,67,021 Cr (8.6% CAGR); Rs 72,398 Cr EBITDA; 5Y CFO/PAT = 1.01x.
- **Two-Stage FCFF DCF Results vs. CMP (Cutoff: Nov 3, 2026;** $R_f = 7.473\%, ERP = 7.08\%, g = 5.0\%$**Rf = 7.473%, ERP = 7.08%, g = 5.0%):**
  - *Trent (10Y Stage 1, WACC 18.34%):* PV Explicit FCFF Rs 6,176.54 Cr + PV TV Rs 6,958.10 Cr = EV Rs 13,134.63 Cr → → **DCF IV = Rs 229.58** vs. CMP Rs 3,036 (**-92.4% Upside / -1,222.4% MoS**).
  - *Asian Paints (5Y Stage 1, WACC 15.06%):* PV Explicit FCFF Rs 12,710.27 Cr + PV TV Rs 25,200.01 Cr = EV Rs 37,910.29 Cr → → **DCF IV = Rs 439.08** vs. CMP Rs 2,438 (**-82.0% Upside**).

- *TCS (5Y Stage 1, WACC 15.20%):* PV Explicit FCFF Rs 1,71,732.69 Cr + PV TV Rs 3,25,281.01 Cr = EV Rs 4,97,013.70 Cr → → **DCF IV = Rs 1,472.12** vs. CMP Rs 2,499 (**-41.1% Upside**).

- **Earnings Quality & Scope Discipline:** 15/15 company-years pass `CFO/PAT > 0.80`. Trent's Rs 2,561.25 Cr debt represents Ind AS 116 store leases. Beneish M-Score and DuPont excluded by scope design.

Slide 5: Valuation Sensitivity & Football-Field Synthesis (3:00 – 3:45)

- 5×5**5 × 5 WACC ×× Terminal Growth Sensitivity Table (`SA_final_workbook.xlsx`):**
  - *Trent:* Ranges from **Rs 195.86** ($WACC = 19.34\%, g = 4.0\%$W ACC = 19.34%, g = 4.0%) to **Rs 274.84** ($WACC = 17.34\%, g = 6.0\%$W ACC = 17.34%, g = 6.0%).
  - *Asian Paints:* Ranges from **Rs 380.89** ($WACC = 16.06\%, g = 4.0\%$ W ACC = 16.06%, g = 4.0%) to **Rs 525.64** ($WACC = 14.06\%, g = 6.0\%$ W ACC = 14.06%, g = 6.0%).
  - *TCS:* Ranges from **Rs 1,274.18** ($WACC = 16.20\%, g = 4.0\%$W ACC = 16.20%, g = 4.0%) to **Rs 1,764.73** ($WACC = 14.20\%, g = 6.0\%$W ACC = 14.20%, g = 6.0%).
- **Relative Valuation Comps (50/50 P/E & EV/EBITDA Blend):**
  - *Trent:* Peer-implied = **Rs 1,540.75** (P/E 40.6x, EV/EBITDA 25.9x); Own 5Y history = **Rs 4,132.13**.
  - *Asian Paints:* Peer-implied = **Rs 1,375.48** (P/E 31.1x, EV/EBITDA 19.7x); Own 5Y history = **Rs 2,679.70** (CMP Rs 2,438 trades at a 9.0% discount to its 5Y median multiples).
  - *TCS:* Peer-implied = **Rs 2,467.90** (CMP Rs 2,499 trades at fair value parity with Infosys/HCL Tech) and a 35.4% discount to Own 5Y history (**Rs 3,869.93**).

Slide 6: AI-Augmented NLP Architecture & Folio (`F.`) AI Strategist (3:45 – 4:30)

- **End-to-End Local Pipeline:**
  1. *Corpus:* 24 Concall Transcripts (8 quarters ×× 3 companies) segmented into **Management Remarks** and **Analyst Q&A** + 150 Financial News Headlines.
  2. *Chunking:* Sentence-preserving 450-token windows respecting transformer token boundaries.
  3. *Dual Sentiment Scoring:* Specialized **ProsusAI FinBERT** vs. open-source local **Ollama Llama 3.1 8B** ($[-1.0, +1.0]$[–1.0, +1.0] score scale).

4. *Live Strategic Co-Pilot (`folio_ai.py`):* Connects the **Folio (`F.`)** Streamlit terminal directly to local `llama3.1:latest` on `http://localhost:11434`, streaming real-time strategic analysis grounded in live DCF/MVO/Ratio state plus on-the-fly custom sentence classification.

Slide 7: Earnings Call Sentiment Results (4:30 – 5:15)

- **Management Optimism Gap:** Management remarks consistently score +0.20 to +0.28 higher than analyst Q&A across all 8 quarters.
- **Company-Specific Sentiment Trajectories:**
  - *Trent:* High positive remarks (+0.70) outpace cautious Q&A (+0.44) as analysts question Tier-3 store economics.
  - *Asian Paints:* Severe sentiment collapse in Q&A from +0.10 (FY25 Q1) down to -0.32 (FY26 Q1) reflecting Birla Opus dealer margin wars.
  - *TCS:* Remarks remain resiliently positive (+0.38) backed by strong quarterly TCV, while Q&A hovers near neutral (+0.10) over GenAI billing rates.

Slide 8: Model Validation & Disagreement Analysis (5:15 – 6:00)

- **90-Sentence Human Gold-Standard Validation:**
  - *FinBERT:* Accuracy = **82.2%**, Macro-F1 = **0.814**, Cohen's Kappa $\kappa = 0.732$κ = 0.732.
  - *Llama 3.1 8B:* Accuracy = **88.9%**, Macro-F1 = **0.885**, Cohen's Kappa $\kappa = 0.829$ κ = 0.829.
- **Why the Local LLM Won:**
  - FinBERT lacks Indian corporate context. When Asian Paints stated *"We calibrated price points selectively to preserve throughput"*, FinBERT labeled it Positive (+0.34); Llama 3.1 and humans correctly labeled it Negative (-0.42), recognizing defensive price discounting.
- **Quote Verification:** 100% of extracted JSON footnotes verified verbatim against source annual reports (0% hallucination rate).

Slide 9: Behavioral Bias & Media Framing Detection (6:00 – 6:45)

- **Classification of 150 Financial Media Headlines (Moneycontrol / ET / LiveMint):**

1. *Trent:* **54% Hype / FOMO framing** ("unstoppable growth engine", "next 100-bagger"), inflating multiple to 94.1x P/E.
2. *Asian Paints:* **58% Extrapolation bias**, treating a cyclical price war as permanent moat destruction.
3. *TCS:* **52% Loss Aversion / Disruption Fear**, discounting the stock to 18.4x P/E despite elite cash generation.

- **The Gap:** Media narratives amplify extremes; fundamental cash flow conversion anchors reality.

Slide 10: Predictive Context: Technicals & Monte Carlo Simulation (6:45 – 7:30)

- **Technical Indicators & Split-Adjusted Feed (`01 / MARKET TERMINAL`):**
  - Automatically detects and adjusts the `2026-01-01` 3:2 share split discontinuity in `TRENT.NS` daily data.
  - Interactive TradingView SPX-style chart with SMA 50/200, Bollinger Bands, RSI (14), and MACD (12, 26, 9).
- **1-Year Geometric Brownian Motion Monte Carlo (1,000 Paths):**
  - Explicitly presented as a **distributional risk scenario, not a point forecast**.
  - Displays 5th, 25th, 50th (Median), 75th, and 95th percentile price cones alongside DCF Intrinsic Value and Peer-Implied reference lines.

Slide 11: Risk & Diversification Architecture (7:30 – 8:15)

- **Portfolio Assets:** 3 Risky Equities (Trent, Asian Paints, TCS) + Sovereign Risk-Free Asset (91-Day RBI T-Bill at 6.80% p.a.).
- **Forward-Looking Expected Returns:**
  - Rather than extrapolating Trent's historical 31.92% run, we use CAPM cost of equity: $E(R_{\text{Trent}}) = 18.51\%, E(R_{\text{AP}}) = 15.24\%, E(R_{\text{TCS}}) = 15.28\%$E(RTrent) = 18.51%, E(RAP) = 15.24%, E(RTCS) = 15.28%.
- **Mean-Variance Optimization (MVO Tangency Portfolio):**
  - **Asian Paints: 49.5% | TCS: 39.4% | Trent: 11.1%** (Constrained by 43.14% vol).
  - Forward Expected Return = **15.62%** | Portfolio Volatility = **21.21%** | Forward Sharpe Ratio = **+0.42**.

Slide 12: Utility Allocation & Risk Questionnaire Engine (8:15 – 9:00)

- **10-Question Questionnaire Structure (`Risk_Aversion_Questionnaire.pdf`):**
  - Section A (Ability / Capacity): Age, Horizon, Income stability, Dependents, Emergency runway.
  - Section B (Willingness / Attitude): Lottery gamble, Career risk, Downturn reaction, Anxiety, Goal.
- **The Prudence Rule:** If $|\text{Sec A} - \text{Sec B}| \geq 6 \implies R_{\text{used}} = 2 \times \min(\text{Sec A}, \text{Sec B})$ |Sec A – Sec B| ≥ 6 ⟹ Rused = 2 × min(Sec A, Sec B) so portfolio risk never exceeds financial capacity.
- **Linear Risk-Aversion Conversion:** $A = 10 - 0.3 \times (R_{\text{used}} - 10)$A = 10 – 0.3 × (Rused – 10).
- **Utility-Optimal Equity Share:** $y^* = \min(1.0, \frac{E(R_p) - R_{f,\text{tbill}}}{A \cdot \sigma_p^2})$y∗ = min (1.0, A ⋅ σp2E(Rp)–Rf,tbill).
  - *Aggressive Basket ($R = 36, A = 2.2$R = 36, A = 2.2):* **89.1% Equity / 10.9% T-Bills**
  - *Balanced Basket ($R = 27, A = 4.9$R = 27, A = 4.9):* **40.0% Equity / 60.0% T-Bills**
  - *Defensive Basket ($R = 15, A = 8.5$R = 15, A = 8.5):* **23.1% Equity / 76.9% T-Bills**

Slide 13: Building the Portfolio Over the Years: 20-Year Monthly SIP (9:00 – 9:45)

- **Dynamic Age-Based Glide Path (`03 / RISK QUESTIONNAIRE & SIP`):**
  - Starting at Rs 25,000/month stepping up 10% annually, modeled deterministically and via a 2,000-path Monte Carlo fan.
  - *Ages 26–32 (Years 1–7):* Aggressive Basket (100% equity). Invested = Rs 28.5L → → Wealth = Rs 47.0L.
  - *Ages 33–40 (Years 8–15):* Balanced Basket (65% equity / 35% T-Bills). Invested = Rs 73.4L → → Wealth = Rs 1.34 Cr.
  - *Ages 41–45 (Years 16–20):* Defensive Basket (35% equity / 65% T-Bills). Total Invested = Rs 1.05 Cr → → **Projected Wealth = Rs 2.18 Crores**.

Slide 14: 5-Year Backtest & Core Performance Ratios (9:45 – 10:30)

- **Evaluation Across the 4 Core Performance Metrics (Oct 2021 – Oct 2026, $R_f = 7.473\%$Rf = 7.473%):**
  1. **Sharpe Ratio:** Trent (**+0.57**), Equal-Weight (**+0.12**) vs. Nifty 50 (**-0.09**), TCS (-0.23), Asian Paints (-0.35).

2. **Treynor's Ratio:** Trent (**+13.33%**), Equal-Weight (**+2.10%**) vs. Nifty 50 (**-1.25%**), TCS (-5.76%), Asian Paints (-7.96%).
3. **Jensen's Alpha (αα):** Trent (**+26.75%**), Equal-Weight (**+4.63% annual abnormal return**) over CAPM.
4. **Information Ratio:** Trent (**+0.69**), Equal-Weight (**+0.25** at 16.69% tracking error).

- **Final Stock Recommendations:**
  - *Trent Ltd.:* **AVOID / PROFIT-TAKING** (Excessive multiple, wide sentiment-to-DCF gap).
  - *Asian Paints Ltd.:* **ACCUMULATE ON DIPS** (Quality franchise de-rated on transient price wars).
  - *TCS Ltd.:* **ACCUMULATE / CORE BALLAST** (Disruption fear masks 1.01x cash conversion and peer multiple parity).

Slide 15: Limitations, Q&A Defense & Live Folio (`F.`) Demo (10:30 – 11:30)

- **Key Academic Limitations:** Historical returns do not predict future performance; DCF terminal value sensitive to Gordon Growth rate; sentiment models carry corporate speech bias.
- **Prepared Q&A Defenses:**
  - *Q: Why TCS over ITC?* Pure-play cash-flow IT allows clean isolation of the GenAI disruption narrative; ITC carries conglomerate tobacco tax distortions.
  - *Q: Why use CAPM expected returns in MVO instead of historical returns?* Historical returns overweight Trent's past momentum (+31.92%) and underweight Asian Paints (-1.64%), leading to hindsight bias and buying at peak multiples.
- **Live Interactive Demo (`http://localhost:8502`):**
  - Show the **Folio (`F.`)** 5-tab terminal: TradingView SPX chart (`01`), MVO & 4-Ratio Matrix (`02`), Live 10-Question Risk Profiler & SIP Glide Path (`03`), Interactive DCF & Concall NLP Lab (`04`), and live **Folio AI Strategist** powered by local Ollama `llama3.1` (`05`).